



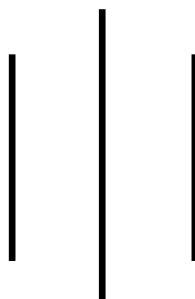
LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal

A Project Proposal

On

“Parking Space Counter”



Submitted to:

Bachelor of Computer Application(BCA) Program

In partial fulfillment of the requirements for the degree of BCA under

1Pokhara University

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ABBREVIATIONS

AD: Advertisement

CRUD: Create Read Update Delete

DOCS: Documents

IDE: Integrated Development Environment

MS: Microsoft

SDLC: Software Development Life Cycle

1. Introduction

The Parking Space Counter is a simple real-life project developed using the C programming language to manage and monitor parking availability in areas such as malls, schools, hospitals, and offices. The main purpose of this project is to count the number of vehicles entering and leaving a parking area and display the available parking spaces. This system helps reduce manual work, saves time, and prevents overcrowding in parking areas. It uses basic programming concepts such as variables, loops, conditions, and functions, making it suitable for beginners who want to learn practical applications of C programming while solving a common real-life problem.

This system helps to increase efficiency by organizing vehicle movement in a systematic way and reducing confusion in parking areas. Similarly, it also gives practical experience in developing management systems that can be applied in smart city and automation projects in the future. Since the project is simple and low-cost, it can be implemented easily in small parking areas without expensive technology. So, for that we developed the system called Parking Space Counter.

For our project, we divided work between us from the beginning. Documentation and coding was divided between all of us. Mainly, Gantt chart was sketched and requirement traceability matrix was done by Sandesh Wagle. Similarly, Waterfall Model was designed by Saksham Giri. Kalyan Karki designed Use Case Diagram. Research about project related information and testing platform and was handled by Arpan Pokhrel. However, each member contributed to the coding and testing of each module and provided their valuable inputs to improve the system.

While doing research we knew that there were many difficulties regarding collection of data, multiple seat booking, data was inaccurate, adaptation in system according to government policy was difficult while changing. By keeping these difficulties we made this system and we were successful to minimize these difficulties using C++.

Going digitally is one of the great idea in this time period because each and every thing has been digitalized and many of the work pressures will be overcome, work will be much faster and we will be able to find each and every thing in much more efficient way. We will be able to deliver a better experience for our customers. Analyzing of data will be much more easier, so going digitally is best method.

Existing system are manual so it is low and also there is high chance of data damage due to external cause, overcrowding was more time consuming. For this environment we developed Parking Space Counter.

2. Problem Statement

There were many difficulties as this was our first programming based project. Although there were many problems we were able to solve and implement it on our project. Manual parking management often causes confusion, errors in counting, and time delays. Drivers may struggle to find available space, and staff may fail to track vehicles properly. This project aims to solve these issues by automating the parking space counter process.

List of the following issues as secondary but prominent issues – can make problem statements in both micro and macro-level.

- Lack of Efficient Parking Space Monitoring
- Absence of Real-Time Information
- Human Error in Manual System
- Poor Utilization of Parking Space
- Time Wastage and Traffic Delay

3. Objectives

The main objective of the Parking Space Counter project is to develop a simple system using the C programming language that can efficiently manage and monitor parking spaces. It aims to automatically track the number of vehicles entering and leaving a parking area and display the available and occupied spaces in real time. This helps to reduce the problems of manual counting, human errors, and confusion in parking management. The project also focuses on improving parking efficiency in places like malls, schools, hospitals, and offices, while saving time for users and ensuring proper use of available parking spaces.

- **To monitor** vehicle entry and exit in the parking area.
- **To calculate** available and occupied parking spaces.
- **To reduce** errors caused by manual counting.
- **To provide** real-time parking status information.
- **To improve** efficiency in parking management systems.
- **To prevent** overcrowding in parking areas.
- **To save** time for drivers and parking staff.
- **To support** better organization of parking facilities.

4. Background Study

The Parking Space Counter project is based on the study of real-life parking management problems faced in busy public and private places such as malls, hospitals, schools, offices, and markets. In these areas, the number of vehicles is increasing rapidly, but parking space is limited. Traditionally, parking is managed manually by attendants who count vehicles entering and exiting the area. This method is time-consuming, error-prone, and often leads to confusion about available parking spaces.

From this situation, it is observed that there is a need for a simple and efficient system to track parking space usage accurately. With the help of basic computer programming, especially C language, it is possible to design a system that can automatically count vehicles and display real-time parking status. The study of this problem shows how automation can improve accuracy, reduce human effort, and save time.

Therefore, this project is developed as a beginner-level software solution that applies fundamental programming concepts like variables, loops, conditions, and functions to solve a practical real-world problem related to parking management.

5. Development

The development phase of **Parking Space Counter** involves a systematic approach to planning, organizing, and executing the technical aspects of the project. This section outlines the Software Development Life Cycle (SDLC) model adopted, the technical environment utilized, and the collaboration strategies employed to ensure the project meets its defined objectives.

5.1 Software Development Life Cycle (SDLC) - Waterfall Model

For this project, we have implemented the Waterfall Model as our primary development paradigm. This model follows a simple step-by-step development process where a simplified version is built, tested, and refined based on feedback before the final product is developed. This model is particularly suitable for **Parking Space Counter** because it is easy to understand, organized, and effective for small projects with clear requirements.

- **Requirement Analysis:** Requirements of the project are collected and studied. The developer identifies the number of parking spaces, vehicle counting method, sensors, display systems, and user needs.
- **System Design:** The overall structure of the system is planned. Circuit diagrams, sensor placement, database design, and software flowcharts are prepared before development begins.
- **Development:** The actual coding and hardware integration are performed. Sensors, microcontrollers, and display units are connected and programmed to count vehicles and show available parking spaces.
- **Testing:** The developed system is tested to check whether it works correctly. Sensor accuracy, counting process, display output, and error handling are verified and bugs are fixed.
- **Deployment:** After successful testing, the parking space counter is installed in the real parking area such as a mall or office parking lot for live operation.

- **Maintenance:** In the final step, the system is monitored and maintained regularly. Faulty components are repaired, software updates are added, and system performances is improved when necessary.

5.2 Development Environment and Tools

The choice of tools was governed by the need for efficiency and the specific technical requirements of the C programming language:

Integrated Development Environment (IDE): We choose code: VS Code because it provides a lightweight environment with a built-in compiler and debugger, which is essential for managing sensors, and microcontroller.

Documentation Tools: Microsoft Word and Google Docs are used to maintain version control over our project report and ensure compliance with formatting guidelines.

5.3 Use Case Description

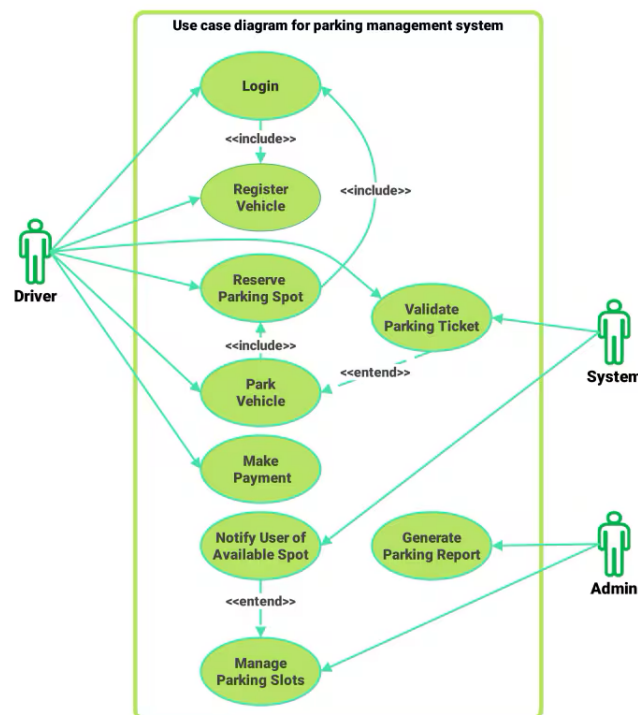


Figure 1: Use case diagram of Parking Space Counter

The Use Case diagram serves as a visual representation of the functional requirements of Parking Space Counter. It illustrates the interactions between the driver (The primary actor) and the various processes within the application.

5.4 Project Timeline (Gantt chart)

A Gantt chart is a project planning tool used to schedule and manage different activities within a specific time period. It helps in organizing tasks step by step and tracking the progress of the project.

For this project, the Gantt chart was used to divide the work into different phases such as planning, system design, coding, testing, and documentation. This helped in completing the project in a systematic and organized manner within the given time.

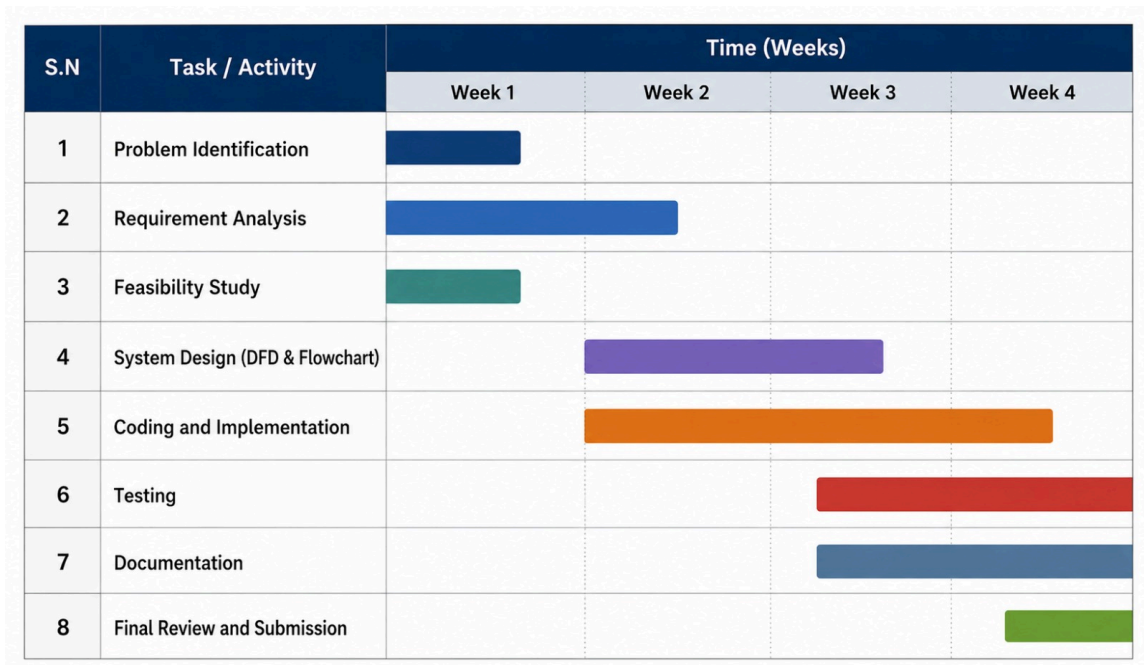


Figure 2: Gantt chart of Parking Space Counter Project activities

6. Project Results

After successful implementation, the parking space counter system works efficiently to detect vehicles entering and leaving the parking area. The sensors accurately count the number of vehicles and continuously update the available parking spaces in real time. The display unit shows the correct number of empty and occupied slots, helping drivers easily find parking spaces without confusion or delay. The system reduces manual work, minimizes traffic congestion inside parking areas, and improves overall parking management, especially in places like shopping malls, hospitals, and offices. Testing results show that the system responds quickly and maintains good accuracy in vehicle detection. Overall, the project result proves that an automated parking space counter is an effective, reliable, and time-saving solution for modern parking management systems.

7. Conclusion

The Parking Space Counter project is a smart and automated system designed to manage parking areas efficiently. It uses sensors to detect vehicles entering and exiting and updates the available parking spaces in real time. After development and testing, the system proved to be reliable, accurate, and useful for reducing manual effort in parking management.

- The system will automatically count vehicles entering and leaving the parking area.
- It will provide real-time information about available parking spaces.
- It will reduce human errors in manual parking management.
- It will help in reducing traffic congestion in busy areas like malls and hospitals.
- It will improve efficiency and save time for both users and management.
- The system is cost-effective and easy to implement.
- It demonstrates the practical use of automation in real-life applications.

8. References

“Embedded Systems: Architecture and Programming” - for microcontroller concepts.

“Electronics Tutorials” - for sensor and circuit understanding.

Udemy (Arduino and IoT projects)